LSST DESC
Dark Energy Science Collaboration

Star-Galaxy Separation with Euclid

Kayleigh Excell, UW Madison

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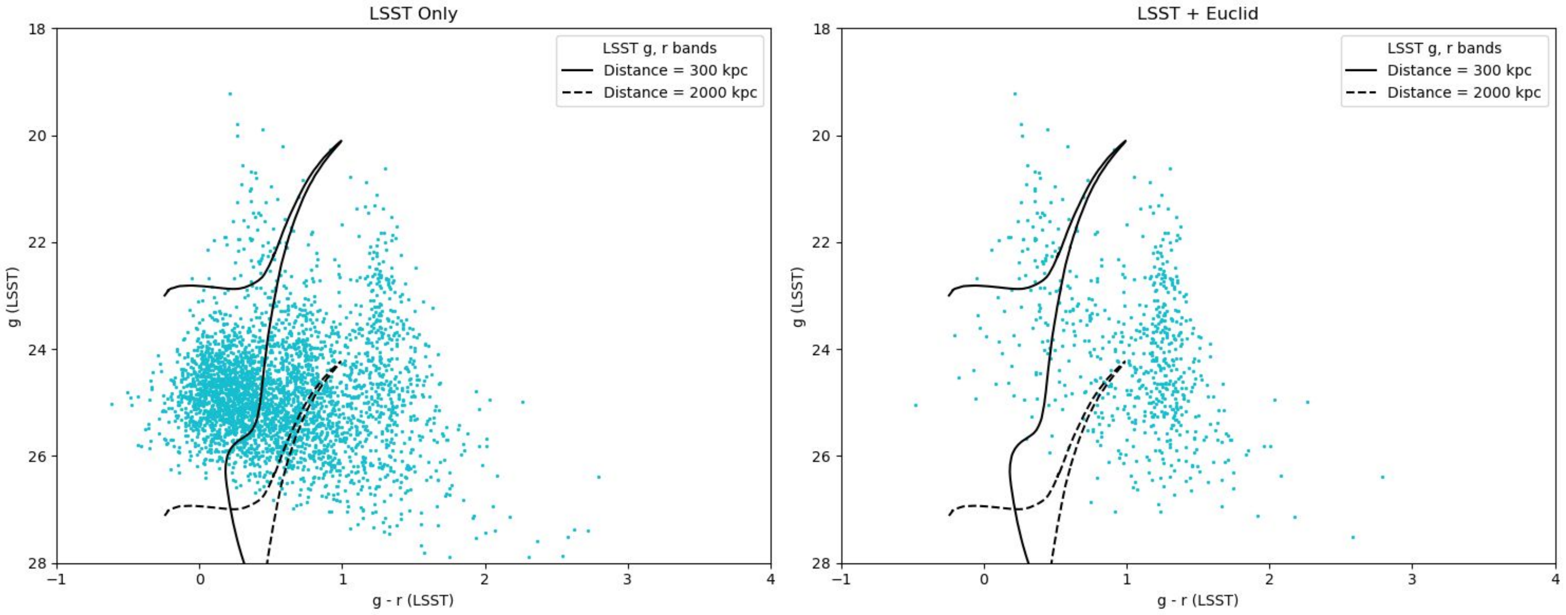
Star-Galaxy Separation with Euclid

Motivation: Euclid's `point_like_prob` can lead to much cleaner separation. We want to optimize using Euclid (or, in the future, Roman) then bring that optimized classifier to other areas of the sky

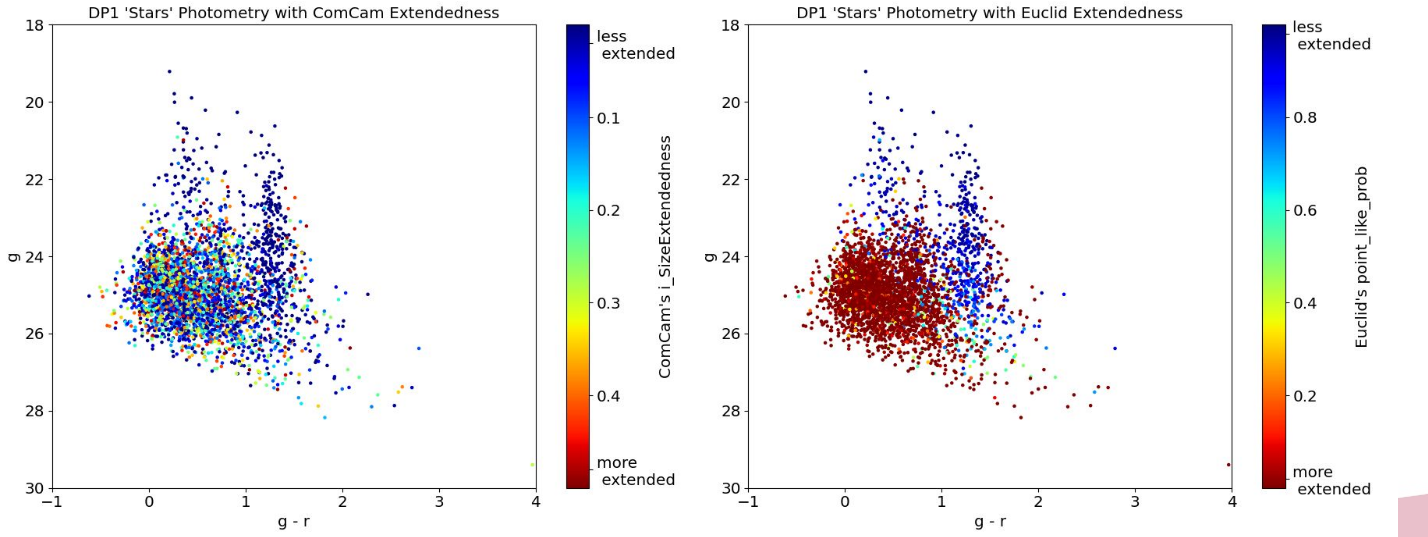


Star-Galaxy Separation in DP1 (EDFS)

For ComCam's stellar sample (selected with `'i_SizeExtendedness'<0.5`)

->Euclid's '`POINT_LIKE_PROB`' shows distinct star and galaxy loci in color-magnitude space better than ComCam's '`i_SizeExtendedness`', especially for faint objects (g > 24)

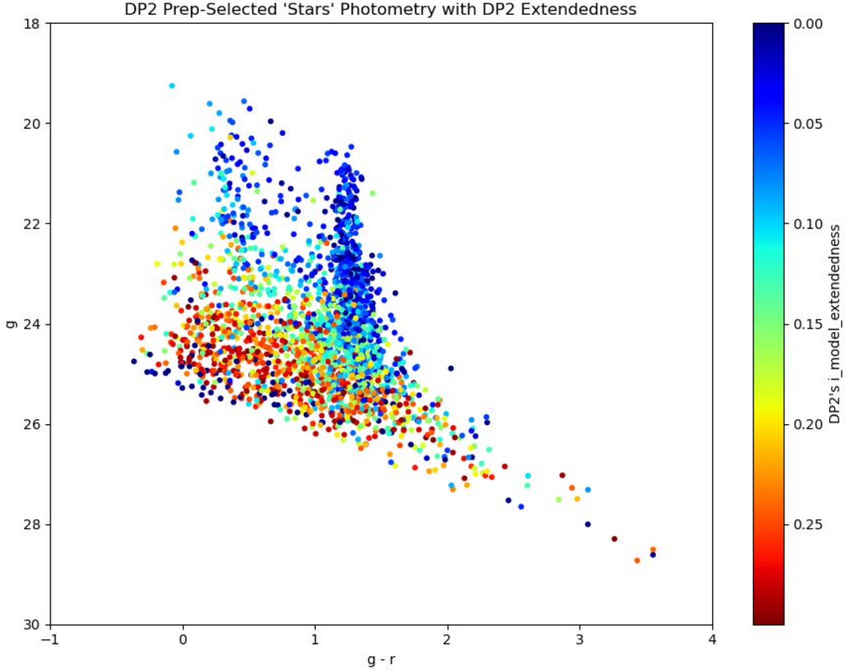
-> Euclid shows that a large fraction of the ComCam-selected "stellar sample" is actually misclassified galaxies, especially for g > 24 & g - r < 1

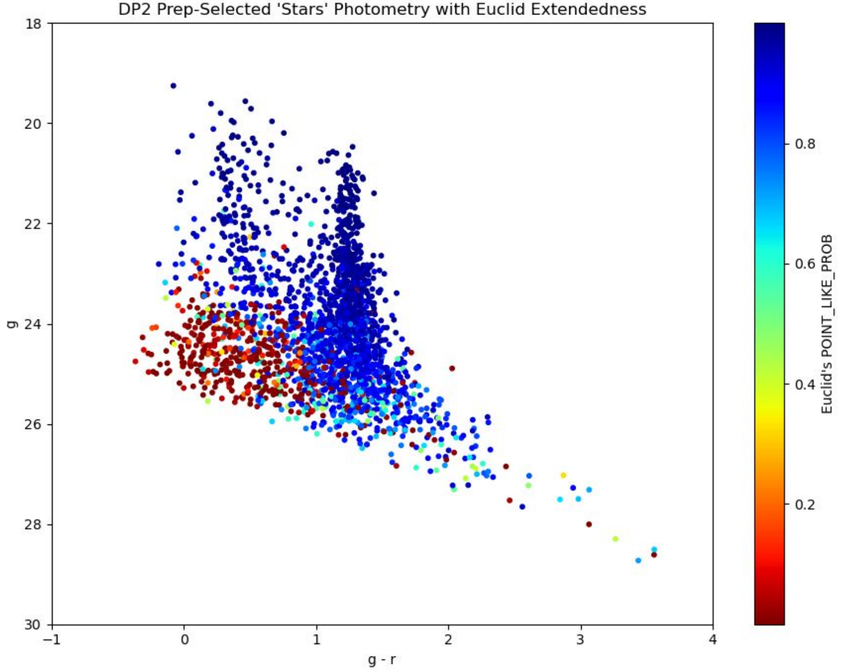


Star-Galaxy Separation in DP2 (EDFS)

DP2 has a new `{band}_model_extendedness`

This is using `i_model_extendedness < 0.3`

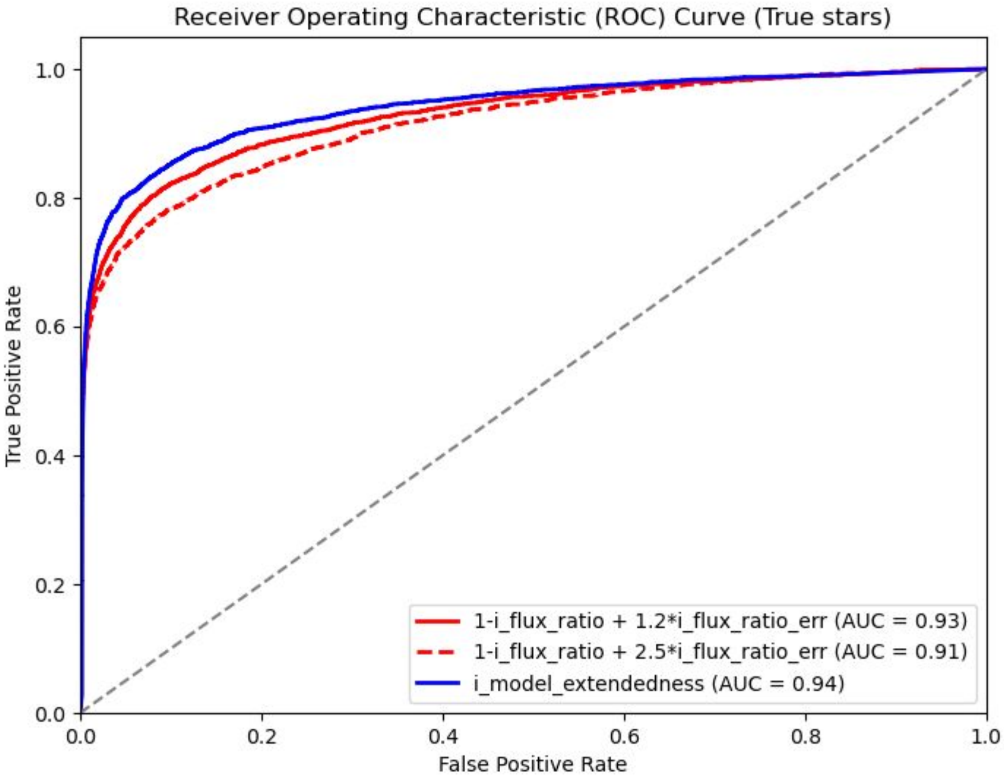




Optimizing Classifiers

Using Euclid as truth labels, we've been testing possible classifiers

This is run on the pre-release DP2 data in tract 2394 (one of the EDFS tracts)



Going Forward

Currently training an undergraduate, Anna Castello, to begin a search Euclid Q1 data, using `point_like_prob>0.5` for her stellar catalog (this is just a Euclid data search)

Euclid DR1 (coming 21 Oct 2026)

Pivoting to getting this working on NERSC (RSP can hold just 1 tract of DP2 data)

More work to be done on characterizing and improving our star-galaxy classifier for DP2

Long term goals this is building to:

Using Euclid DR1 and LSST DR1 to search for and characterize UFDs and stellar streams -> not just s-g separation but quantifying sensitivity using overlapping regions

Extra Plots

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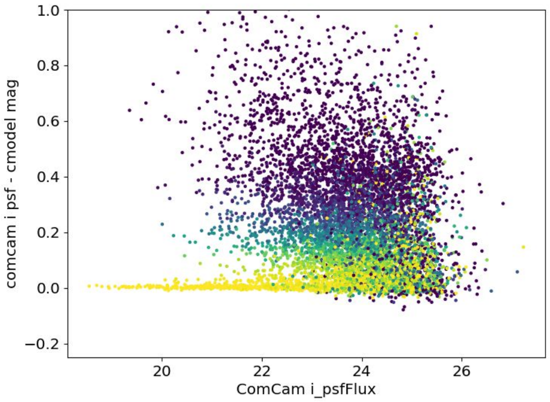
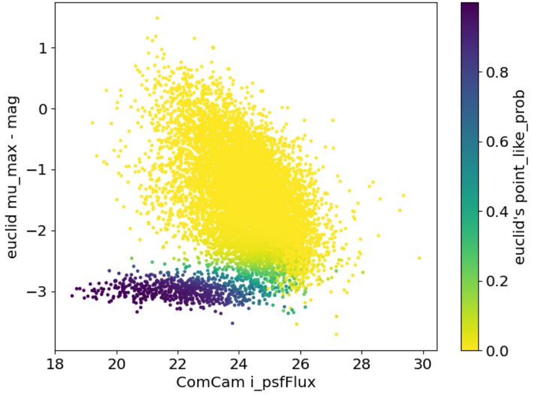
DP1:





DP2: